

Project Design Phase-II Data Flow Diagram & User Stories

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Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagram (DFD) Specifications Report

Overview

The **Data Flow Diagram (DFD Level-1)** represents the structural architectural mechanics and information flow for the **Automated Network Request Management System** in ServiceNow. It tracks data ingestion from initial submission to verification, approval orchestration, fulfillment task creation, and permanent database commit. **Core Elements of DFD Level-1**

1. External Entities (Actors)

- **Employee (Requester):** Initiates data ingestion workflows via Service Portal interactions (submits network requests, views real-time status updates, receives notifications).
- **Line Manager (Approver):** Authorizes or rejects execution path changes (reviews request parameters, approves/rejects, adds comments).
- **Network Administrator (Fulfillment):** Executes physical or logical network provisioning tasks (receives approved work orders, performs installation, updates asset inventory).

2. Processes (System Transformations)

- **Process 1.0 (Network Request Submission):** Front-end UI application captures request details, uploaded network architecture documents, and user login tokens.

- **Process 2.0 (Form Validation & User Verification):** Checks mandatory field inputs, executes IPv4 Regex validation, and auto-populates employee profile variables.
- **Process 3.0 (Approval Workflow - Flow Designer):** Executes business validation state gates, creates approval records (sysapproval_approver), and routes tasks to the designated Line Manager.
- **Process 4.0 (Request Processing):** Spawns catalog fulfillment tasks (sc_task), notifies the Network Operations Group, and updates request state to *In-Progress*.
- **Process 5.0 (Inventory & Database Update):** Commits approved values to permanent datastores and updates physical infrastructure inventory records.
- **Process 6.0 (Notification & Request Closure):** Issues completion notifications to the requester and closes out the parent requested item (sc_req_item).

3. Data Stores (Persistence Ledger)

- **D1: sys_user (User Table):** Core source profile repository storing employee names, email addresses, department assignments, phone numbers, and role permissions.
- **D2: u_network_database (Custom Database Table):** The ultimate permanent tracking schema ledger storing request details, hardware/device information, network port configurations, building locations, approval logs, and installation completion timestamps.

Execution Flow Summary

1. **Request Ingestion:** The Employee submits a network request through the Service Portal interface.
2. **Profile Verification:** Process 2.0 queries sys_user (D1) to auto-populate and verify employee credentials.
3. **Approval Orchestration:** Process 3.0 triggers Flow Designer, logging a pending state in u_network_database (D2) while awaiting Line Manager authorization.
4. **Task Execution:** Upon approval, Process 4.0 creates an sc_task assigned to the Network Administrator.
5. **Database Commit:** Process 5.0 writes final installation and port configuration details to u_network_database (D2).
6. **Closure:** System sends confirmation notifications to the requester and closes the ticket.

AUTOMATED NETWORK REQUEST DFD

AUTOMATED NETWORK REQUEST MANAGEMENT

EXTERNAL ENTITIES

EMPLOYEE (REQUESTER)

- Submit Network Request
- View Request Status
- Receive Notifications

LINE MANAGER (APPROVER)

- Review Request
- Approve / Reject
- Add Comments

NETWORK ADMINISTRATOR

- Receive Assignment Tasks
- Update & Install Network
- Network Inventory Management

PROCESSES

1.0 NETWORK REQUEST SUBMISSION

- Submit Network Request
- Upload Required Documents

Submitted Data

2.0 FORM VALIDATION & USER VERIFICATION

- Validate Form Input (IPv4 Regex)
- Verify & Auto-populate User Profile

Validated Data

3.0 APPROVAL WORKFLOW (FLOW DESIGNER)

- Review Approval Log
- Log / Approval History
- Log Request Progress

Validated Data

4.0 REQUEST PROCESSING

- Configure & Review Network
- Install / Update Network
- Install / Update Service
- Manage Request Status

Processed Data

5.0 INVENTORY & DATABASE UPDATE

- Update Network Inventory
- Store Request Details
- Update Request Details

DATA STORES

D1

sys_user (User Table)

Stores

- Employee Profile
- Department
- Contact Number
- User Tables
- User Roles

D2

u_network_database
(Custom Database)

Stores

- Request Details
- Inventory / Device
- Approval History
- Request Status Details
- Completion Date

External Entity Process Data Store → Data Flow → Conditional Flow

User Stories Table

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Requester (Portal User)	Catalog Submission	USN-1	As a requester, I want to submit a network request via Service Portal with autovalidated fields (e.g., IPv4 format).	Form validates IPv4 inputs and hides irrelevant fields dynamically based on request type.	High	Sprint-1
Requester (Portal User)	Status Visibility	USN-2	As a requester, I want to view real-time status updates and SLA progress on my submitted RITM record.	Portal displays current state (Awaiting Approval, In Progress, Completed) and active SLA timers.	High	Sprint-1

Approver (Manager / Security)	Approval Routing	USN-3	As an approver, I want to receive dynamic email notifications with one-click approval buttons.	Approvals route conditionally based on request priority, access level, and security risk.	High	Sprint-1
Network Engineer	Task Fulfillment	USN-4	As a network engineer, I want auto-generated sc_task records populated with complete technical parameters.	sc_task is created immediately upon approval with target CI details mapped.	High	Sprint-2
User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Administrator	CMDB & SLA Tracking	USN-5	As an admin, I want completed network changes to auto-update CMDB records and track SLA breaches on dashboards.	cmdb_ci_netgear updates upon task completion and SLA metrics display on executive reports.	Medium	Sprint-2

